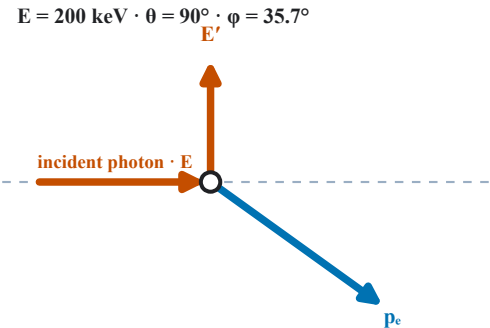


Compton scattering · exact relativistic kinematics

BPhO Computational Challenge 2026 · Task 9 · five official incident energies · stationary free electron

44/44 kinematic checks · 30/30 extension checks

Exact 200 keV collision at $\theta = 90^\circ$ · momentum-vector construction



Fractional wavelength shift

0.39139

$\Delta\lambda = 2.426 \text{ pm}$

Electron recoil speed

0.43419 c

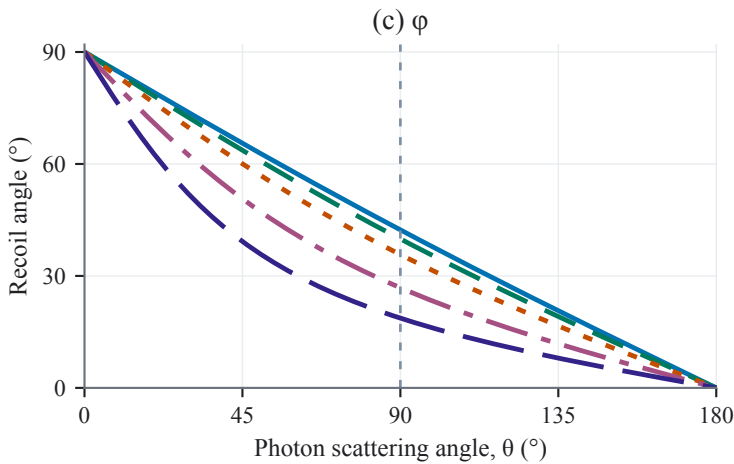
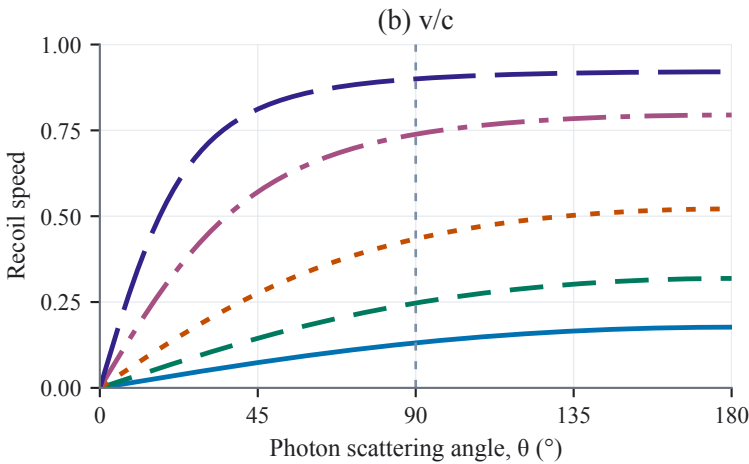
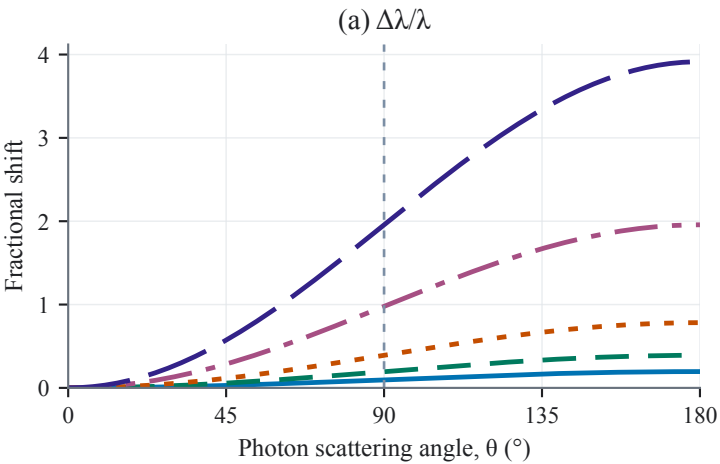
fully relativistic

Electron recoil angle

35.71°

below incident axis

$E' = 143.741 \text{ keV} \cdot K = 56.259 \text{ keV}$



$\theta = 0^\circ$: zero electron momentum and undefined φ (the plots show its 90° limit) · $\theta = 180^\circ$: maximum shift and $\varphi = 0^\circ$

50 keV · solid 100 keV · dashed 200 keV · dotted 500 keV · dash-dot 1000 keV · long dash